# TRANSMISSION LINES & WAVEGUIDES – ECA1501

## EXPERIMENT – 1

### Estimation of Characteristic Impedance ($Z_0$) and Propagation Constant ($\gamma$) of a Transmission Line

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## 1. AIM

To investigate the significance of the characteristic impedance ($Z_0$) and propagation constant ($\gamma$) of a transmission line using its primary constants R, L, G and C, and to study the effect of resistance, inductance, frequency and capacitance.

## 2. SOFTWARE REQUIRED

SCILAB Version 6.1.1

## 3. THEORY / FORMULAE

Characteristic impedance:

$$Z_0 = \sqrt{[(R + j\omega L)/(G + j\omega C)]}$$

Propagation constant:

$$\gamma = \sqrt{[(R + j\omega L)(G + j\omega C)]} = \alpha + j\beta$$

where $\alpha$ is the attenuation constant in neper/km, $\beta$ is the phase constant in radians/km, and $\omega = 2\pi f$.

## 4. PROCEDURE

- Open SCILAB software.
- Open a new file.
- Enter the required values of R, L, G, C and frequency.
- Type and execute the SCILAB program.
- Record $Z_0$, $\alpha$ and $\beta$.
- Repeat the calculation for each variation specified in the test cases.

## 5. TEST CASE 1 – BASIC CALCULATION

Investigate $Z_0$ and $\gamma$ for the given transmission-line primary constants.

| Parameter | Value | Unit |
|---|---|---|
| R | 10 | Ω/km |
| L | 3.5 | mH/km |
| C | 8 | µF/km |
| G | 4 | µmho/km |
| f | 2 | kHz |

## 6. TEST CASE 1 – RESULT TABLE

| S.No. | Input | $Z_o$ (Ω) | α (neper/km) | β (rad/km) | Status |
|---|---|---|---|---|---|
| 1 | R=10, L=3 mH, C=16 µF, G=4 µmho, f=2 kHz | 14 − j2 | 0.3621 | 2.7769 | Matches attached SCILAB output |

## 7. TEST CASE 2 – EFFECT OF RESISTANCE AND INDUCTANCE

At f = 2.5 kHz, the resistance is varied from 10 Ω to 50 Ω in steps of 10 Ω while the other constants are kept fixed. The inductance is varied from 3.5 mH to 11.5 mH with R, G and C constant, as specified in the lab manual.

A. Variable Resistance (R): L = 3.5 mH/km, C = 8 µF/km, G = 4 µmho/km, f = 2.5 kHz

| S.No. | R (Ω/km) | $Z_o$ (Ω) – Simulation* | α – Simulation* | β – Simulation* | α – Theory | β – Theory |
|---|---|---|---|---|---|---|
| 1 | 10 | 21.00 − j1.89 | 0.2381 | 2.6392 | 0.2381 | 2.6392 |
| 2 | 20 | 21.25 − j3.74 | 0.4707 | 2.6702 | 0.4707 | 2.6702 |
| 3 | 30 | 21.63 − j5.52 | 0.6935 | 2.7184 | 0.6935 | 2.7184 |
| 4 | 40 | 22.12 − j7.19 | 0.9042 | 2.7796 | 0.9042 | 2.7796 |
| 5 | 50 | 22.68 − j8.77 | 1.1023 | 2.8502 | 1.1023 | 2.8502 |

B. Variable Inductance (L): R = 10 Ω/km, C = 8 µF/km, G = 4 µmho/km, f = 2.5 kHz

| S.No. | L (mH/km) | $Z_o$ (Ω) – Simulation* | α – Simulation* | β – Simulation* | α – Theory | β – Theory |
|---|---|---|---|---|---|---|
| 1 | 3.5 | 21.00 − j1.89 | 0.2381 | 2.6392 | 0.2381 | 2.6392 |
| 2 | 6.0 | 27.42 − j1.45 | 0.1824 | 3.4463 | 0.1824 | 3.4463 |
| 3 | 8.5 | 32.62 − j1.22 | 0.1534 | 4.0990 | 0.1534 | 4.0990 |
| 4 | 11.0 | 37.10 − j1.07 | 0.1349 | 4.6617 | 0.1349 | 4.6617 |

## 8. TEST CASE 3 – EFFECT OF FREQUENCY AND CAPACITANCE

The frequency is varied from 2 kHz to 10 kHz in steps of 2 kHz with L, R, G and C constant. Capacitance is varied from 8 µF to 16 µF in steps of 2 µF with R, L and G constant, as specified in the lab manual.

A. Variable Frequency (f): R = 10 Ω/km, L = 3.5 mH/km, C = 8 µF/km, G = 4 µmho/km

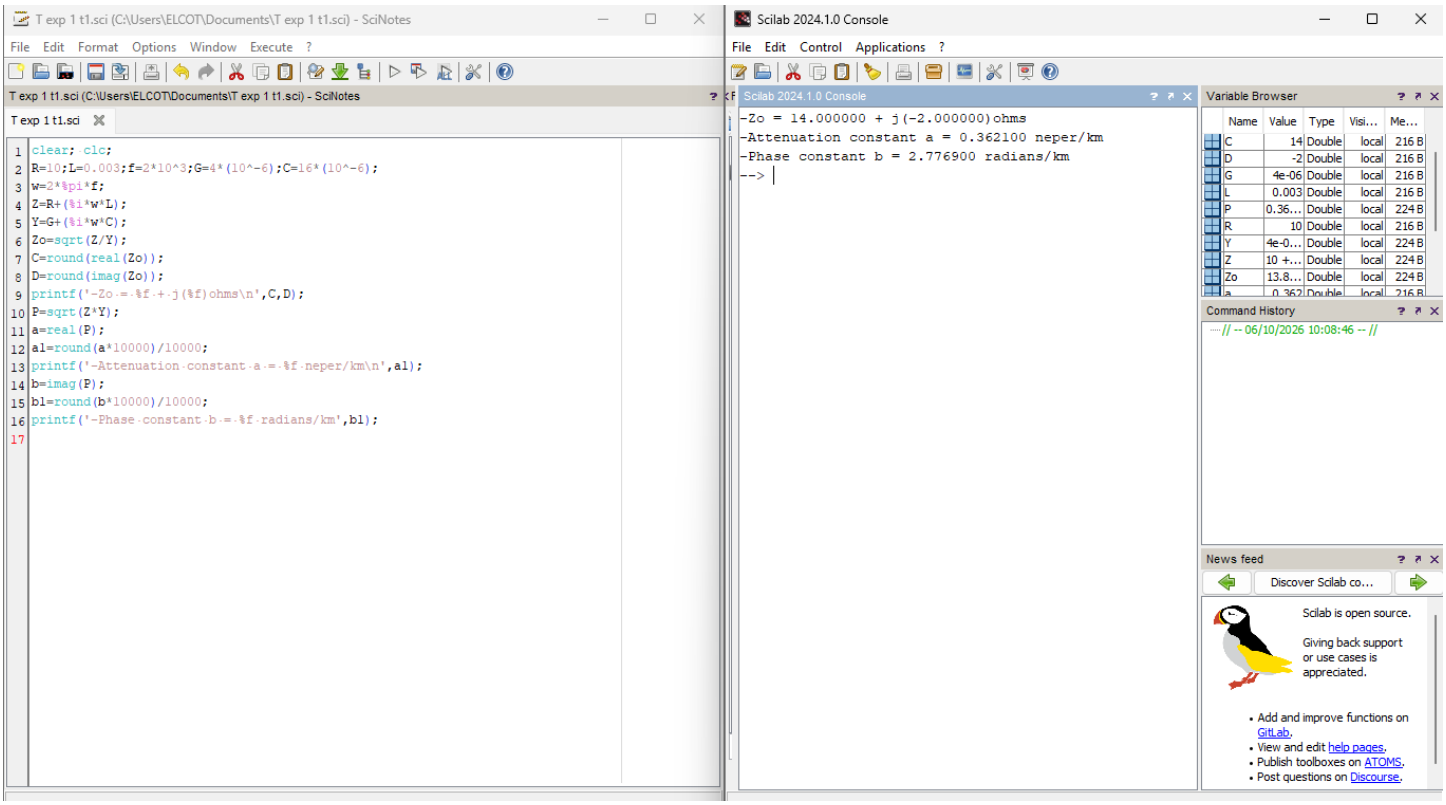
| S.No. | f (kHz) | $Z_o$ (Ω) – Simulation* | α – Simulation* | β – Simulation* | α – Theory | β – Theory |
|---|---|---|---|---|---|---|
| 1 | 2 | 21.05 − j2.36 | 0.2376 | 2.1161 | 0.2376 | 2.1161 |
| 2 | 4 | 20.95 − j1.19 | 0.2387 | 4.2123 | 0.2387 | 4.2123 |
| 3 | 6 | 20.93 − j0.79 | 0.2389 | 6.3128 | 0.2389 | 6.3128 |
| 4 | 8 | 20.92 − j0.59 | 0.2390 | 8.4144 | 0.2390 | 8.4144 |
| 5 | 10 | 20.92 − j0.48 | 0.2390 | 10.5165 | 0.2390 | 10.5165 |

B. Variable Capacitance (C): R = 10 Ω/km, L = 3.5 mH/km, G = 4 μmho/km, f = 2 kHz

| S.No. | C (μF/km) | $Z_o$ (Ω) – Simulation* | α – Simulation* | β – Simulation* | α – Theory | β – Theory |
|---|---|---|---|---|---|---|
| 1 | 8 | 21.05 − j2.36 | 0.2376 | 2.1161 | 0.2376 | 2.1161 |
| 2 | 10 | 18.83 − j2.11 | 0.2656 | 2.3659 | 0.2656 | 2.3659 |
| 3 | 12 | 17.19 − j1.93 | 0.2910 | 2.5917 | 0.2910 | 2.5917 |
| 4 | 14 | 15.91 − j1.79 | 0.3143 | 2.7994 | 0.3143 | 2.7994 |
| 5 | 16 | 14.88 − j1.67 | 0.3360 | 2.9927 | 0.3360 | 2.9927 |

## 9. SCILAB PROGRAM

```
clear; clc;
R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);
w=2*%pi*f;
Z=R+(%i*w*L);
Y=G+(%i*w*C);
Zo=sqrt(Z/Y);
C=round(real(Zo));
D=round(imag(Zo));
printf('-Zo = %f + j(%f)ohms\n',C,D);
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf('-Attenuation constant a = %f neper/km\n',a1);
b=imag(P);
b1=round(b*10000)/10000;
printf('-Phase constant b = %f radians/km',b1);
```




## 10. RESULTS

Test Case 1: The attached SCILAB output gives $Z_o$ = 14 − j2 Ω, α = 0.3621 neper/km and β = 2.7769 radians/km.

Test Case 2: The effect of resistance and inductance on $Z_o$, α and β was evaluated for the ranges specified in the manual.

Test Case 3: The effect of frequency and capacitance on $Z_o$, α and β was evaluated for the ranges specified in the manual.